Curiosity Rover Spots Mysterious Cliffs Ahead on Mars

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NASA's Curiosity **rover** has sent back a beautiful new **panorama** of Mars that gives scientists their clearest look yet at a line of strange cliffs shaped by wind.

The **rover** took six photos with its **Mastcam** camera on August 11, at 8:30 in the morning Mars time. Scientists joined them into one wide view. They chose not to adjust the colors, so the picture keeps the real look of a Martian morning, with bluish tones in front and sunlight on the cliffs in the distance.

These cliffs are called **yardangs**. They are ridges carved by wind, and they stretch about 10 miles across part of Mount Sharp, a mountain about 3 miles tall. Curiosity has been climbing Mount Sharp since 2014. It recently gained about 0.6 miles in height, a bigger climb than any other **rover** on Mars.

Mount Sharp is made of many layers of rock, and each layer holds clues about a different time in Mars's past. These layers have shown that billions of years ago there were lakes and streams in this area. When the water dried up, salty minerals were left behind.

But the yardang layer is puzzling. Its color is different from the rocks around it, and its layers tilt at an odd angle. "It almost appears plastered on," said Ashwin Vasavada, a Curiosity scientist at NASA's Jet Propulsion Laboratory. He said one idea is that it may be ash from ancient volcanic **eruptions**.

Over the next year, Curiosity will keep driving through rocks full of minerals that formed as Mars dried out. Scientists hope it reaches the bottom of the cliffs in 2027, where it can study them with its **robotic arm** and help solve the mystery.

Word Treasure

panorama -- A wide picture that shows a whole scene, often made by joining several photos.

yardangs -- Long ridges or cliffs of rock carved by wind.

Mastcam -- The main color camera on the Curiosity rover's tall mast.

rover -- A robot vehicle that drives around on another planet or moon.

eruptions -- Times when a volcano bursts out lava, ash or gas.

robotic arm -- A machine arm on the rover that can reach out and use tools on rocks.

Background you need

Curiosity is a car-sized NASA rover exploring Gale Crater on Mars, where it has been climbing Mount Sharp since 2014.

Mount Sharp is built from many rock layers, and each layer works like a page in a history book that records what Mars was like long ago.

By studying these layers, scientists learned that the area once had lakes and streams before Mars became the dry, cold planet it is today.

Rovers carry cameras and tools on robotic arms so scientists on Earth can study rocks up close without traveling to Mars.

Break it down

LEAD: Curiosity sends back its clearest view yet of wind-carved cliffs on Mars
HOW THE PICTURE WAS MADE: Six Mastcam photos from August 11, colors left unadjusted
PLACE: Yardangs stretching about 10 miles across Mount Sharp
+- MILESTONE: Curiosity has climbed about 0.6 miles, more than any other rover
+- HISTORY: Rock layers show ancient lakes and streams that later dried up
MYSTERY: The yardang layer has an odd color and tilt; one idea is volcanic ash
EXPERT: Ashwin Vasavada of NASA's Jet Propulsion Laboratory
NEXT STEPS: Drive through mineral-rich layers and reach the cliffs in 2027
OPEN QUESTION: What really created the yardang layer?

Why it matters

Each rock layer Curiosity studies helps explain how Mars changed from a world with lakes and streams to the dry planet we see today.

Test what you remember

1. What are the wind-carved cliffs in the panorama called?

- (A) Volcanoes
- (B) Canyons
- (C) Yardangs
- (D) Craters

2. How many photographs were combined to make the panorama?

- (A) Two
- (B) Twelve
- (C) One hundred
- (D) Six

3. Since what year has Curiosity been climbing Mount Sharp?

- (A) 2014
- (B) 2027
- (C) 2024
- (D) 2020

4. What do Mount Sharp's rock layers show about Mars long ago?

- (A) It was covered in ice
- (B) It had lakes and streams
- (C) It had forests
- (D) It had oceans of lava only

5. What is one idea for what formed the unusual yardang layer?

- (A) Dust from a passing comet
- (B) Snow from a recent storm
- (C) Ice from the polar caps
- (D) Ash from ancient volcanic eruptions

6. When do scientists hope Curiosity will reach the bottom of the cliffs?

- (A) In 2014
- (B) Never
- (C) Next week
- (D) In 2027

Answer key (try the quiz first!): 1: C 2: D 3: A 4: B 5: D 6: D

Questions to think about

1. Ashwin Vasavada, Curiosity scientist at NASA's Jet Propulsion Laboratory: Says the yardang layer looks out of place and may be ash from ancient volcanic eruptions, which makes it exciting to reach.

MY THOUGHTS (at least 20 words):